



Up in the air

Powered flight took off in 1903 when American brothers Wilbur and Orville Wright attached an engine to a glider and traveled through air for 12 seconds. This short flight blazed the trail for supersonic jets, giant airliners, and even spacecraft.

1913 Russian Pyotr Nesterov becomes the first pilot to fly a loop-the-loop.



1785 Frenchman Jean-Pierre Blanchard and American John Jeffries fly across the English Channel in a balloon.



1783

In Paris, France, the Montgolfier brothers' hot-air balloon makes the world's first manned flight, lasting 25 minutes.

1900 The first rigid airship, the Zeppelin LZ1, makes its maiden voyage in Germany.

1903

The Wright brothers' first powered flying machine, the *Wright Flyer*, takes off in the USA.

1850



1852

Frenchman Jules Henri Giffard's steam-powered airship makes its first flight, proving controlled flight is possible.

1900

1907

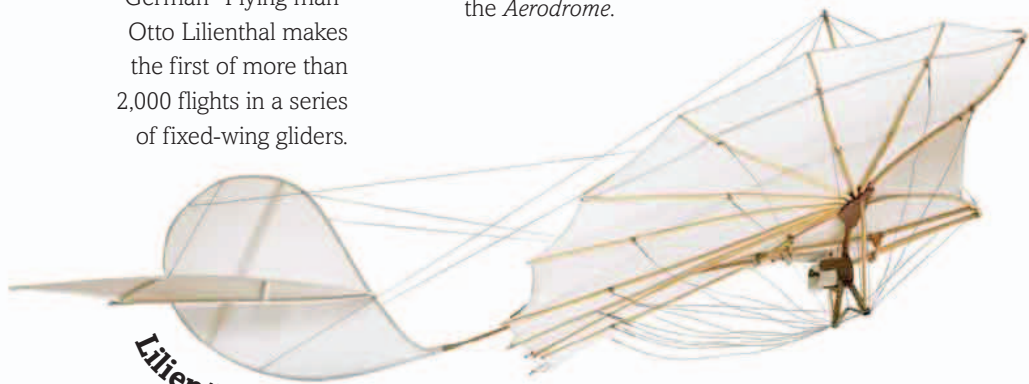
The first flight of a rotary-wing aircraft, forerunner of the helicopter, is piloted by French engineer Paul Cornu.

1896

American inventor Samuel Pierpont Langley flies his steam-powered model aircraft, the *Aerodrome*.

1891

German "Flying man" Otto Lilienthal makes the first of more than 2,000 flights in a series of fixed-wing gliders.



Lilienthal's Normal Apparatus